Answer Key: Topological Basics and Continuity

Mathematical Foundations for Machine Learning

1 Open and Closed Sets

Exercise 1.1 ($\star$). *Let $\Omega = (0,1) \subset \mathbb{R}$. Using the definition of an open set, prove that Ω is open.*

Answer

Let $x \in (0,1)$. Choose $\varepsilon = \min(x, 1-x) > 0$. Then for any z with $|z - x| < \varepsilon$, we have $z > x - \varepsilon \geq x - x = 0$ and $z < x + \varepsilon \leq x + (1-x) = 1$. Thus $B(x,\varepsilon) = (x-\varepsilon, x+\varepsilon) \subset (0,1)$, so $(0,1)$ is open.

Exercise 1.2 ($\star$). *Determine whether the following subsets of $\mathbb{R}^2$ are open, closed, both, or neither.*

Answer

(a) $A = \{(x,y) : x^2+y^2 < 1\}$: **Open**. For any $p \in A$ with $\|p\|_2 = r < 1$, take $\varepsilon = 1-r > 0$. Then $B(p,\varepsilon) \subset A$.

(b) $B = \{(x,y) : x^2+y^2 \leq 1\}$: **Closed**. Its complement $\{(x,y) : x^2+y^2 > 1\}$ is open (for any p with $\|p\| = r > 1$, take $\varepsilon = r-1$).

(c) $C = \{(x,y) : 0 < x^2+y^2 \leq 1\}$: **Neither**. Not open because $(1,0) \in C$ but every ball around $(1,0)$ contains points with $\|(x,y)\| > 1$. Not closed because the sequence $(1/n, 0) \in C$ converges to $(0,0) \notin C$.

(d) $D = \mathbb{R}^2$: **Both open and closed** (clopen). Open by definition; closed because $\mathbb{R}^2 \setminus \mathbb{R}^2 = \emptyset$ is open.

Exercise 1.3 ($\star\star$). *Prove that the open first quadrant $\Omega = \{(x,y) \in \mathbb{R}^2 : x > 0, y > 0\}$ is open.*

Answer

Let $(x,y) \in \Omega$, so $x > 0$ and $y > 0$. Choose $\varepsilon = \min(x,y) > 0$. For any (z_1,z_2) with $\|(z_1,z_2) - (x,y)\|_2 < \varepsilon$, we have $|z_1 - x| < \varepsilon \leq x$, so $z_1 > 0$. Similarly $|z_2 - y| < \varepsilon \leq y$, so $z_2 > 0$. Thus $B((x,y),\varepsilon) \subset \Omega$.

Exercise 1.4 ($\star\star$). *Using the sequential characterization, prove that $[0,1]$ is closed.*

Answer

Let (x_k) be a convergent sequence with $x_k \in [0,1]$ for all k, and let $x = \lim_{k\to\infty} x_k$. Since $0 \leq x_k \leq 1$ for all k, taking limits preserves non-strict inequalities: $0 \leq x \leq 1$ (if $x < 0$, then for large k, $x_k < 0$, contradiction; similarly for $x > 1$). Thus $x \in [0,1]$, and $[0,1]$ is closed.

Exercise 1.5 ($\star\star\star$). *Prove properties of intersections/unions of open sets. Give a counterexample for infinite intersections.*

Answer

Finite intersection: Let $U_1, \ldots, U_m$ be open and $x \in \bigcap_{i=1}^m U_i$. For each i, there exists $\varepsilon_i > 0$ with $B(x, \varepsilon_i) \subset U_i$. Take $\varepsilon = \min(\varepsilon_1, \ldots, \varepsilon_m) > 0$ (finite minimum of positive numbers). Then $B(x, \varepsilon) \subset U_i$ for all i, so $B(x, \varepsilon) \subset \bigcap_{i=1}^m U_i$.
Arbitrary union: Let $\{U_\alpha\}_{\alpha \in I}$ be open sets and $x \in \bigcup_\alpha U_\alpha$. Then $x \in U_\beta$ for some β, so there exists $\varepsilon > 0$ with $B(x, \varepsilon) \subset U_\beta \subset \bigcup_\alpha U_\alpha$.
Counterexample: $\bigcap_{n=1}^\infty (-1/n, 1/n) = \{0\}$, which is not open.

2 Closure, Interior, and Boundary

Exercise 2.1 ($\star$). *For $\Omega = (0, 1]$, find $\overline{\Omega}$, Ω^o, $\partial\Omega$.*

Answer

$\overline{\Omega} = [0, 1]$ (since 0 is a limit point: $1/n \to 0$ with $1/n \in \Omega$).
$\Omega^o = (0, 1)$ (the point 1 has no open ball contained in Ω).
$\partial\Omega = \{0, 1\}$.
Verification: $\overline{\Omega} \setminus \Omega^o = [0, 1] \setminus (0, 1) = \{0, 1\} = \partial\Omega$. ✓

Exercise 2.2 ($\star$). *For $\Omega = \mathbb{Q} \cap [0, 1]$, find $\overline{\Omega}$, Ω^o, $\partial\Omega$.*

Answer

$\overline{\Omega} = [0, 1]$, because every real number in $[0, 1]$ is a limit of rationals (density of $\mathbb{Q}$).
$\Omega^o = \emptyset$, because every interval contains irrationals, so no open ball is contained in Ω.
$\partial\Omega = [0, 1] \setminus \emptyset = [0, 1]$.

Exercise 2.3 ($\star\star$). *For $\Omega = \{(x, y) : x^2 + y^2 < 1\} \cup \{(1, 0)\}$, find closure, interior, and boundary.*

Answer

$\overline{\Omega} = \{(x, y) : x^2 + y^2 \leq 1\}$ (the closed unit disk, since all boundary points of the open disk are limit points).
$\Omega^o = \{(x, y) : x^2 + y^2 < 1\}$ (the isolated point $(1, 0)$ has no open ball contained in Ω since every neighborhood of $(1, 0)$ contains points outside the closed disk).
$\partial\Omega = \{(x, y) : x^2 + y^2 = 1\}$ (the unit circle).

Exercise 2.4 ($\star\star$). *Prove that Ω^o is open, $\overline{\Omega}$ is closed, and Ω is open iff $\Omega = \Omega^o$.*

Answer

(a) Let $x \in \Omega^o$. Then $\exists\, \varepsilon > 0$ with $B(x, \varepsilon) \subset \Omega$. For any $y \in B(x, \varepsilon)$, let $\delta = \varepsilon - \|y - x\| > 0$. Then $B(y, \delta) \subset B(x, \varepsilon) \subset \Omega$, so $y \in \Omega^o$. Thus $B(x, \varepsilon) \subset \Omega^o$.
(b) We show $\mathbb{R}^n \setminus \overline{\Omega}$ is open. If $x \notin \overline{\Omega}$, then $\exists\, \varepsilon > 0$ with $B(x, \varepsilon) \cap \Omega = \emptyset$. For $y \in B(x, \varepsilon)$, take $\delta = \varepsilon - \|y - x\|$; then $B(y, \delta) \cap \Omega = \emptyset$, so $y \notin \overline{\Omega}$.
(c) ($\Rightarrow$) If Ω is open, every $x \in \Omega$ has a ball inside Ω, so $x \in \Omega^o$, giving $\Omega \subset \Omega^o$. Since $\Omega^o \subset \Omega$ always, $\Omega = \Omega^o$. ($\Leftarrow$) If $\Omega = \Omega^o$, then Ω is open by part (a).

Exercise 2.5 ($\star\star\star$). *Prove or disprove the identities for closures and interiors of unions/intersections.*

Answer

(a) **True.** ($\supset$): $A \subset A \cup B \Rightarrow \overline{A} \subset \overline{A \cup B}$; similarly for B. ($\subset$): $\overline{A} \cup \overline{B}$ is closed and contains $A \cup B$, so $\overline{A \cup B} \subset \overline{A} \cup \overline{B}$.

(b) **False.** Let $A = (0, 1)$ and $B = (1, 2)$ in $\mathbb{R}$. Then $A \cap B = \emptyset$, so $\overline{A \cap B} = \emptyset$, but $\overline{A} \cap \overline{B} = [0, 1] \cap [1, 2] = \{1\}$.

(c) **True.** ($\subset$): $A \cap B \subset A$ and $A \cap B \subset B$, so $(A \cap B)^o \subset A^o$ and $(A \cap B)^o \subset B^o$. ($\supset$): $A^o \cap B^o$ is open (intersection of two open sets) and $A^o \cap B^o \subset A \cap B$, so $A^o \cap B^o \subset (A \cap B)^o$.

(d) **False.** Let $A = [0, 1]$ and $B = [1, 2]$ in $\mathbb{R}$. Then $A^o \cup B^o = (0, 1) \cup (1, 2)$, but $(A \cup B)^o = (0, 2)$, which contains 1.

3 Boundedness, Compactness, and Bolzano–Weierstrass

Exercise 3.1 ($\star$). *Determine which sets are bounded.*

Answer

(a) $A = \{x : \|x\|_2 \leq 5\}$: **Bounded**, contained in $B(0, 6)$.

(b) $B = \{x : x_1^2 - x_2^2 < 1\}$: **Not bounded**. For any t, $(0, t) \in B$ since $-t^2 < 1$.

(c) $C = \{x : \|x\|_1 < 10\}$: **Bounded**. Since $\|x\|_2 \leq \|x\|_1 < 10$, we have $C \subset B_2(0, 10)$.

Exercise 3.2 ($\star$). *Determine which sets are compact.*

Answer

(a) $[0, 1]^n$: **Compact**. Closed and bounded ($\subset B(0, \sqrt{n})$).

(b) $(0, 1)^n$: **Not compact**. Not closed (e.g., $(1/k, 1/2, \ldots, 1/2) \to (0, 1/2, \ldots, 1/2) \notin (0, 1)^n$).

(c) Unit sphere $\{x : \|x\|_2 = 1\}$: **Compact**. Closed (preimage of $\{1\}$ under the continuous map $x \mapsto \|x\|$) and bounded.

(d) $\{1/n : n \in \mathbb{N}\}$: **Not compact**. Not closed: $1/n \to 0 \notin \{1/n\}$.

(e) $\{0\} \cup \{1/n : n \in \mathbb{N}\}$: **Compact**. Bounded ($\subset [0, 1]$) and closed (the only limit point 0 is included).

Exercise 3.3 ($\star\star$). *Show that $x_k = (\cos(k), (-1)^k/k)$ is bounded and apply Bolzano–Weierstrass.*

Answer

We have $|\cos(k)| \leq 1$ and $|(-1)^k/k| \leq 1$, so $\|x_k\|_\infty \leq 1$ for all k. Thus (x_k) is contained in $[-1, 1]^2$, which is bounded. By the Bolzano–Weierstrass theorem, every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence, so (x_k) has a convergent subsequence.

Exercise 3.4 ($\star\star$). *Prove: if every convergent subsequence of $(x_k) \subset K$ converges to x^*, then $x_k \to x^*$.*

Answer

Suppose for contradiction that $x_k \not\to x^*$. Then there exists $\varepsilon > 0$ and a subsequence (x_{k_j}) with $\|x_{k_j} - x^*\| \geq \varepsilon$ for all j. Since K is compact, (x_{k_j}) has a further convergent subsequence $(x_{k_{j_l}})$ converging to some $y \in K$. This is a convergent subsequence of the original sequence, so by assumption $y = x^*$. But $\|x_{k_{j_l}} - x^*\| \geq \varepsilon$ for all l, which contradicts $x_{k_{j_l}} \to x^*$.

Exercise 3.5 ($\star\star\star$). *Nested compact sets and counterexample.*

Answer

Proof: For each i, pick $x_i \in K_i$. Since $K_i \subset K_1$ for all i (nested), the sequence (x_i) lies in K_1, which is compact. By Bolzano–Weierstrass, there exists a convergent subsequence $x_{i_j} \to x^*$ with $x^* \in K_1$. For any fixed N, all terms x_{i_j} with $i_j \geq N$ satisfy $x_{i_j} \in K_N$ (since $K_{i_j} \subset K_N$). As K_N is closed, $x^* \in K_N$. This holds for all N, so $x^* \in \bigcap_{i=1}^{\infty} K_i$.
Counterexample: $K_i = [i, \infty)$ in $\mathbb{R}$. Each K_i is closed and nested, but $\bigcap_{i=1}^{\infty} K_i = \emptyset$ (no real number is $\geq i$ for all i). These sets are not bounded, hence not compact.

4 Continuity

Exercise 4.1 ($\star$). *Prove $f(x) = 3x + 2$ is continuous using ε-δ.*

Answer

Let $\varepsilon > 0$ and $x_0 \in \mathbb{R}$. Choose $\delta = \varepsilon/3$. Then $|x - x_0| < \delta$ implies

$$|f(x) - f(x_0)| = |3x + 2 - 3x_0 - 2| = 3|x - x_0| < 3\delta = \varepsilon.$$

Thus f is continuous at every x_0.

Exercise 4.2 ($\star$). *Prove that $f(x) = \|x\|_2$ is continuous.*

Answer

By the reverse triangle inequality, $\big|\|x\|_2 - \|x_0\|_2\big| \leq \|x - x_0\|_2$. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $\|x - x_0\|_2 < \delta$ implies $|f(x) - f(x_0)| \leq \|x - x_0\|_2 < \varepsilon$.

Exercise 4.3 ($\star\star$). *Use the preimage characterization to show the open/closed unit ball.*

Answer

Since $f(x) = \|x\|_2$ is continuous:

- $\{x : \|x\|_2 < 1\} = f^{-1}((-\infty, 1))$. The set $(-\infty, 1)$ is open in $\mathbb{R}$, so by Proposition 3, $f^{-1}((-\infty, 1))$ is open.
- $\{x : \|x\|_2 \leq 1\} = f^{-1}((-\infty, 1])$. The set $(-\infty, 1]$ is closed in $\mathbb{R}$, so $f^{-1}((-\infty, 1])$ is closed.

Exercise 4.4 ($\star\star$). *Is the continuous image of an open set necessarily open?*

Answer

No. Counterexample: let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$. Then $U = (-1, 1)$ is open, but $f(U) = [0, 1)$ which is not open (there is no open ball around 0 contained in $[0, 1)$).

Exercise 4.5 ($\star\star\star$). *Prove uniform continuity on compact sets.*

Answer

Suppose f is not uniformly continuous on K. Then there exists $\varepsilon_0 > 0$ and sequences $(x_k), (y_k) \in K$ with $\|x_k - y_k\| < 1/k$ but $|f(x_k) - f(y_k)| \geq \varepsilon_0$. Since K is compact, (x_k) has a convergent subsequence $x_{k_j} \to x^* \in K$. Then $y_{k_j} \to x^*$ as well (since $\|y_{k_j} - x_{k_j}\| < 1/k_j \to 0$). By continuity of f at x^*:

$$|f(x_{k_j}) - f(y_{k_j})| \leq |f(x_{k_j}) - f(x^*)| + |f(x^*) - f(y_{k_j})| \to 0,$$

contradicting $|f(x_{k_j}) - f(y_{k_j})| \geq \varepsilon_0$.

5 Extreme Value Theorem and Distance to Sets

Exercise 5.1 ($\star$). *Find min and max of $f(x) = x^2 - 2x + 3$ on $[0, 3]$.*

Answer

$f'(x) = 2x - 2 = 0 \Rightarrow x = 1$. Evaluating: $f(0) = 3$, $f(1) = 2$, $f(3) = 6$.
Minimum: $f(1) = 2$. Maximum: $f(3) = 6$.
Existence guaranteed by Corollary 7: f is continuous and $K = [0, 3]$ is compact (closed and bounded in $\mathbb{R}$).

Exercise 5.2 ($\star\star$). *Find min/max of $f(x) = x_1^2 + 2x_2^2 - x_1$ on the closed unit disk.*

Answer

K is compact (closed and bounded), f is continuous, so Corollary 7 guarantees min and max exist.
Interior critical points: $\nabla f = (2x_1 - 1, 4x_2) = 0$ gives $x_1 = 1/2, x_2 = 0$. This is in K since $\|(1/2, 0)\| < 1$. Value: $f(1/2, 0) = 1/4 - 1/2 = -1/4$.
Boundary ($x_1^2 + x_2^2 = 1$): parametrize $x_1 = \cos\theta, x_2 = \sin\theta$. Then $g(\theta) = \cos^2\theta + 2\sin^2\theta - \cos\theta = 1 + \sin^2\theta - \cos\theta$. Setting $g'(\theta) = 2\sin\theta\cos\theta + \sin\theta = \sin\theta(2\cos\theta + 1) = 0$ gives $\theta = 0, \pi, 2\pi/3, 4\pi/3$. Values: $g(0) = 0$, $g(\pi) = 2$, $g(2\pi/3) = g(4\pi/3) = 9/4$.
Minimum: $f(1/2, 0) = -1/4$. **Maximum:** $f(-1/2, \pm\sqrt{3}/2) = 9/4$.

Exercise 5.3 ($\star\star$). *Show the triangle K is compact, find $d(p, K)$ for $p = (2, 2)$.*

Answer

K is the intersection of three closed half-planes: $\{x_1 \geq 0\}$, $\{x_2 \geq 0\}$, $\{x_1 + x_2 \leq 1\}$. Hence K is closed. It is bounded since $0 \leq x_1, x_2 \leq 1$. Thus K is compact.
The closest point in K to $p = (2, 2)$ is found by projecting onto the hypotenuse $x_1 + x_2 = 1$ (with $x_1, x_2 \geq 0$). By symmetry of $p = (2, 2)$ about the line $x_1 = x_2$, the closest point is $m^* = (1/2, 1/2)$.
$d(p, K) = \|(2, 2) - (1/2, 1/2)\|_2 = \|(3/2, 3/2)\|_2 = \frac{3\sqrt{2}}{2}$.

Exercise 5.4 ($\star\star\star$). *Prove the distance to a closed set is attained.*

Answer

Pick any $m_0 \in M$ and let $r = \|x - m_0\|$. Any minimizer must satisfy $\|x - u\| \leq r$, so it lies in $\overline{B(x,r)} \cap M$. This set is bounded (contained in $\overline{B(x,r)}$) and closed (intersection of closed sets). Hence $K := \overline{B(x,r)} \cap M$ is compact and non-empty ($m_0 \in K$). The function $g(u) = \|x - u\|$ is continuous on K. By Corollary 7, g attains its minimum on K at some $m^* \in K \subset M$. Since any $u \in M \setminus K$ satisfies $\|x - u\| > r \geq g(m^*)$, we have $g(m^*) = d(x, M)$.

6 Path-Connectedness

Exercise 6.1 ($\star$). *Show every convex set is path-connected.*

Answer

Let $x, y \in C$. Define $\gamma : [0,1] \to C$ by $\gamma(t) = (1-t)x + ty$. By convexity, $\gamma(t) \in C$ for all $t \in [0,1]$. Also $\gamma(0) = x$ and $\gamma(1) = y$, and γ is continuous (it is affine). Thus C is path-connected.

Exercise 6.2 ($\star\star$). *Show $\mathbb{R}^2 \setminus \{(0,0)\}$ is path-connected.*

Answer

Let $x, y \in \mathbb{R}^2 \setminus \{0\}$. If the line segment from x to y does not pass through the origin, use $\gamma(t) = (1-t)x + ty$.
If it does (i.e., $y = -\alpha x$ for some $\alpha > 0$), pick any z not on the line through x and the origin (e.g., a perpendicular vector). Then connect x to z and z to y via straight lines; neither passes through the origin. Concatenating these paths gives a continuous curve in $\mathbb{R}^2 \setminus \{0\}$.

Exercise 6.3 ($\star\star$). *Prove continuous image of path-connected set is path-connected.*

Answer

Let $a, b \in f(M)$. Then $a = f(x)$, $b = f(y)$ for some $x, y \in M$. Since M is path-connected, there exists a continuous curve $\gamma : [0,1] \to M$ with $\gamma(0) = x$, $\gamma(1) = y$. Define $\tilde{\gamma} = f \circ \gamma : [0,1] \to f(M)$. Then $\tilde{\gamma}$ is continuous (composition of continuous functions), $\tilde{\gamma}(0) = f(x) = a$, and $\tilde{\gamma}(1) = f(y) = b$. Thus $f(M)$ is path-connected.

Exercise 6.4 ($\star\star\star$). $\mathrm{GL}_n(\mathbb{R})$ *is open with two path-connected components.*

Answer

(a) $\det : \mathbb{R}^{n\times n} \to \mathbb{R}$ is a polynomial in the matrix entries, hence continuous. $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$. Since $\mathbb{R} \setminus \{0\}$ is open and det is continuous, the preimage is open by Proposition 3.
(b) $\mathrm{GL}_n^+ = \det^{-1}((0,\infty))$ and $\mathrm{GL}_n^- = \det^{-1}((-\infty,0))$ are both open, and they partition $\mathrm{GL}_n(\mathbb{R})$. No continuous path can go from GL_n^+ to GL_n^-: if $\gamma(t)$ were such a path, then $\det(\gamma(t))$ would be a continuous function changing sign without hitting zero, contradicting the intermediate value theorem. Each component is path-connected: any matrix in GL_n^+ can be connected to I via a continuous path (e.g., through row reduction operations that preserve positive determinant).

7 Mixed Challenging Exercises

Exercise 7.1 ($\star\star\star$). *Prove sublevel sets of continuous functions are closed.*

Answer

$S_c = \{x : f(x) \leq c\} = f^{-1}((-\infty, c])$. Since $(-\infty, c]$ is closed in $\mathbb{R}$ and f is continuous, the preimage is closed by Proposition 3.
Relevance to optimization: In constrained optimization $\min f(x)$ s.t. $g_i(x) \leq 0$, the feasible set $\bigcap_i \{x : g_i(x) \leq 0\}$ is an intersection of closed sets (hence closed) when the g_i are continuous. If additionally bounded, the feasible set is compact, guaranteeing existence of a minimizer by Corollary 7.

Exercise 7.2 ($\star\star\star$). *Prove compact sets have finite diameter attained.*

Answer

Finite: K compact $\Rightarrow$ K bounded $\Rightarrow$ $K \subset B(0, r)$ for some r. For any $x, y \in K$: $\|x - y\| \leq \|x\| + \|y\| \leq 2r$. So $\operatorname{diam}(K) \leq 2r < \infty$.
Attained: Define $g : K \times K \to \mathbb{R}$ by $g(x, y) = \|x - y\|$. Then g is continuous, and $K \times K$ is compact (product of compact sets). By Corollary 7, g attains its maximum on $K \times K$. This maximum equals $\operatorname{diam}(K)$.

Exercise 7.3 ($\star\star\star$). *Uniform limit of continuous functions is continuous; pointwise counterexample.*

Answer

Proof: Let $\varepsilon > 0$. By uniform convergence, $\exists N$ such that $\sup_{x \in K} |f_N(x) - f(x)| < \varepsilon/3$. Since f_N is continuous at x_0, $\exists \delta > 0$ with $|f_N(x) - f_N(x_0)| < \varepsilon/3$ when $\|x - x_0\| < \delta$. Then:

$$|f(x) - f(x_0)| \leq \underbrace{|f(x) - f_N(x)|}_{<\varepsilon/3} + \underbrace{|f_N(x) - f_N(x_0)|}_{<\varepsilon/3} + \underbrace{|f_N(x_0) - f(x_0)|}_{<\varepsilon/3} < \varepsilon.$$

Counterexample: On $K = [0, 1]$, let $f_k(x) = x^k$. Then $f_k \to f$ pointwise where $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. This f is discontinuous at $x = 1$. The convergence is not uniform since $\sup_{x \in [0,1)} |x^k| = 1$ for all k.

Exercise 7.4 ($\star\star\star$). *Properties of linear maps $f(x) = Ax$.*

Answer

(a) *Continuous:* $\|f(x) - f(y)\| = \|A(x - y)\| \leq \|A\| \, \|x - y\|$ where $\|A\|$ is the operator norm (finite for any matrix). So f is Lipschitz continuous.

(b) *Compact image:* Follows from Theorem 53: continuous image of a compact set is compact.

(c) *Path-connected image:* Follows from Proposition 4: continuous image of a path-connected set is path-connected.

(d) $\ker(A)$ *closed:* $\ker(A) = f^{-1}(\{0\})$. Since $\{0\}$ is closed in $\mathbb{R}^m$ and f is continuous, the preimage is closed by Proposition 3.

Exercise 7.5 ($\star\star\star$). ***(Application to ML)*** *Neural network layer* $f(x) = \sigma(Wx + b)$.

Answer

(a) The affine map $g(x) = Wx + b$ is continuous (as shown in the previous exercise, linear maps are Lipschitz; adding a constant preserves continuity). Since σ is continuous by assumption, $f = \sigma \circ g$ is the composition of continuous functions, hence continuous.

(b) Since K is compact and f is continuous, by Theorem 53, $f(K)$ is compact – hence closed and bounded. This means the output of the layer lies in a bounded region of $\mathbb{R}^m$. By induction, a multi-layer network maps compact inputs to compact outputs, which is useful for stability analysis.

(c) If the set of global minima of the loss function (over weight space) is path-connected, gradient-based methods can potentially navigate between any two minima via continuous paths of non-increasing loss. If the minima split into disconnected components, the initialization determines which component is reachable. This is analogous to $\mathrm{GL}_n(\mathbb{R})$ having two components: algorithms that preserve the sign of the determinant cannot cross between them. Understanding the topology of the loss landscape thus informs whether gradient descent can find all optimal solutions.